Data Communication and Networking

Mid Term Examination Solutions (E1PY206T)

Section A: Network Basics & Topology

1. Define the working of a Gateway in a Computer Network.

A Gateway is a network node that connects two networks using different protocols or architectures. It acts as a protocol converter, translating data from one stack to another (e.g., connecting an OSI-based network to a TCP/IP network). It operates at the higher layers of the OSI model, typically the Application layer.

2. Explain the advantages of Star Topology.

In a Star Topology, all devices are connected to a central hub or switch.

- **Centralized Control:** Management and monitoring are easier through the central hub.
- **Fault Isolation:** If one link fails, only that specific node is affected; the rest of the network remains functional.
- **Scalability:** It is easy to add or remove devices without disrupting the entire network.

3. Differentiate between Simplex, Half-Duplex, and Full-Duplex.

Mode	Description	Example
Simplex	One-way communication only.	Television Broadcast
Half-Duplex	Two-way communication, but not simultaneously.	Walkie-Talkie
Full-Duplex	Simultaneous two-way communication.	Telephone Call

Section B: Transmission & Encoding

4. Explain any two types of Unguided Media.

Unguided media provide a means for transmitting electromagnetic waves without using a physical conductor.

- **Radio Waves:** Omnidirectional waves used for multicast communications like AM/FM radio and cordless phones.
- **Microwaves:** Unidirectional waves that require line-of-sight propagation, used for satellite and cellular communication.

5. Apply encoding techniques for the bit stream: 11010.

- **Manchester Encoding:** A transition occurs at the middle of each bit period. A '1' is typically a high-to-low transition and '0' is a low-to-high transition.
- **Differential Manchester:** A transition at the beginning of a bit interval represents '0', while the absence of a transition represents '1'.

Section C: Design & Conversion

6. Evaluate the different types of computer networks (LAN, MAN, WAN).

- **LAN (Local Area Network):** Privately owned, covering a small area like an office or home. High speed, low delay.
- **MAN (Metropolitan Area Network):** Covers a city-wide area (e.g., cable TV networks).
- **WAN (Wide Area Network):** Provides long-distance transmission over countries or continents (e.g., the Internet).

7. Evaluate the complete Pulse Code Modulation (PCM) process.

PCM is used to convert analog signals into digital form:

1. **Sampling:** Measuring the signal amplitude at regular intervals (Nyquist rate).
2. **Quantization:** Rounding the sampled values to discrete levels.
3. **Encoding:** Converting the quantized levels into binary bits.